IIT JAMMU SUMMER SCHOOL 2026

IoT & Drones Program

Assignment: Fundamentals of Drones and UAV Systems

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Q1. What is a drone (UAV)? Definition, history, and evolution. (5 Marks)

Ans: Definition: A drone (also called a UAV, short for Unmanned Aerial Vehicle) is simply a flying machine with no pilot sitting inside it. Nobody sits inside to fly it - instead, a person flies it from the ground using a remote control, or it can even fly on its own using a built-in computer and sensors. In simple words, it's a flying robot.

History and Evolution: Drones are not a brand-new idea - people have been trying to build pilotless flying machines for a very long time. Here is how they slowly developed over the years:

- 1849 - One of the earliest examples: Austria attached explosives to unmanned balloons and used them against the city of Venice.
- 1917 - The USA built the 'Kettering Bug', an early pilotless flying bomb. It never actually got used in a war, but it proved the idea could work.
- 1930s-40s - Countries started building simple radio-controlled drones (like the British 'Queen Bee') mainly to use as flying targets for training soldiers.
- 1980s-90s - Militaries began using drones like the 'Predator' for spying and watching enemy areas from the sky, instead of just as targets.
- 2006 onwards - Governments started allowing normal people and companies to legally fly drones, not just the military.
- 2010s-Present - Once small sensors, GPS chips, cameras, and batteries became cheap, drones became something anyone could buy and fly - farmers, photographers, students, and even delivery companies started using them.

So basically, drones started as big, expensive military machines, and over time became small, smart, and affordable gadgets that regular people use today.

Q2. Real-world applications of drones. (5 Marks)

Ans: Agriculture: Drones spray pesticides, monitor crop health using multispectral cameras, and estimate yield, saving farmers time and water.

Delivery services: Companies like Amazon and Zomato/Swiggy have tested drone delivery for fast last-mile delivery of parcels and food.

Disaster management: Drones survey flood or earthquake affected areas, locate trapped people using thermal cameras, and drop emergency supplies.

Filmmaking and photography: Aerial shots for movies, weddings, and sports events are captured using camera-equipped drones instead of expensive cranes or helicopters.

Surveillance and defence: Military and police use drones for border monitoring, reconnaissance, and crowd surveillance.

Infrastructure inspection: Drones inspect power lines, bridges, and wind turbine blades without needing a person to climb to dangerous heights.

Q3. Differentiate between the following. (10 Marks)

Ans: (a) UAV vs Drone

Technically, UAV (Unmanned Aerial Vehicle) is the correct, formal engineering term for any aircraft without an onboard human pilot. 'Drone' is the informal, popular name for the same thing. So every drone is a UAV, and in everyday conversation both words are used interchangeably, but 'UAV' is preferred in technical or defence documentation while 'drone' is used in common speech.

Point of Difference	UAV	Drone
Full form / meaning	Unmanned Aerial Vehicle	No official full form - just a common nickname
Nature of term	Formal, technical/engineering term	Informal, everyday term
Where it's used	Defence, aviation authorities, research papers	Common speech, media, hobbyist/commercial talk
Scope	Covers all unmanned aircraft, including large military and fixed-wing types	Usually pictured as a small multirotor, though it means the same as UAV
Example of use	'The UAV completed its reconnaissance mission'	'I flew my drone to take some photos'

(b) Fixed-wing vs Multirotor drones

Feature	Fixed-wing Drone	Multirotor Drone
Structure	Looks like a small airplane with fixed wings	Has multiple rotors/propellers (usually 4, 6 or 8)
Lift generation	Forward motion over wings generates lift	Motors and propellers directly generate lift
Flight time	Longer (can fly for hours)	Shorter (typically 20-40 minutes)
Hovering	Cannot hover in one place	Can hover perfectly in one spot
Use case	Mapping large farmland, long-distance survey	Photography, inspection, delivery in tight spaces

(c) Quadcopter vs Hexacopter

Feature	Quadcopter	Hexacopter
Number of rotors	4 rotors	6 rotors
Stability	Good, but a single motor failure usually causes a crash	Better, since it can often still fly safely if one motor fails
Payload capacity	Lower	Higher, good for carrying cameras or heavy sensors
Cost & complexity	Cheaper and simpler to build/maintain	More expensive, heavier, more wiring and ESCs

Q4. How a quadcopter flies. (10 Marks)

Ans: A quadcopter flies by carefully balancing four forces: lift, gravity, thrust, and drag.

- Gravity constantly pulls the drone down towards the earth.
- The four propellers spin and push air downward. By Newton's third law, the air pushes back on the propellers upward - this reaction force is called lift or thrust. When total lift equals gravity, the drone hovers; when lift is greater, it climbs; when lift is less, it descends.
- Drag is the air resistance that opposes the drone's motion through the air, acting opposite to its direction of travel.

Why CW and CCW motors? Every propeller, as it spins, also creates a small twisting reaction force on the drone body in the opposite direction (like how a helicopter needs a tail rotor to stop spinning). If all four propellers spun the same way, the whole drone body would spin uncontrollably. To cancel this out, two propellers spin clockwise (CW) and the two diagonally opposite ones spin counter-clockwise (CCW). Their rotational effects cancel each other, so the drone body stays stable and only moves the way the pilot commands. Changing the relative speed of CW motors versus CCW motors is also how the drone controls yaw (rotation).

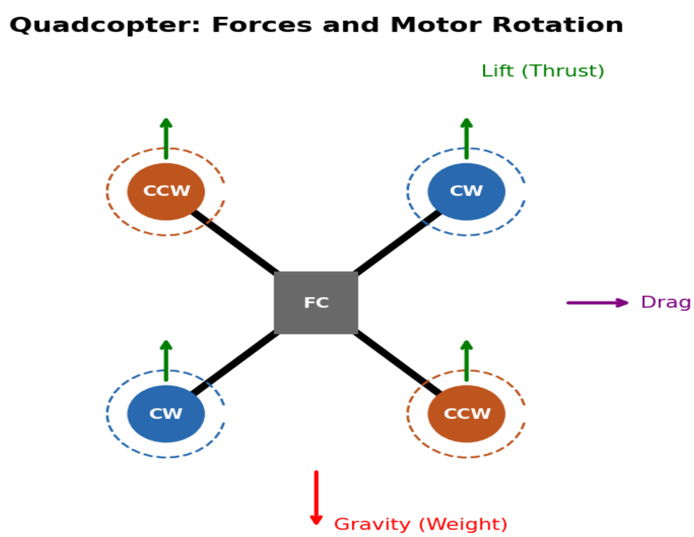

Figure 1: Forces on a quadcopter and CW/CCW motor arrangement

Q5. Take-off, landing, hovering, and directional movement. (10 Marks)

Ans:

- Take-off: All four motors increase speed equally, so total lift becomes greater than gravity, and the drone rises straight up.
- Hovering: All four motors spin at the same, steady speed so that lift exactly equals gravity. The drone stays still in the air at a constant altitude.
- Landing: All four motors reduce speed together so that lift becomes slightly less than gravity, and the drone descends gently until it touches the ground.

- Move forward: The two rear motors spin slightly faster than the two front motors. This tilts (pitches) the nose of the drone down, and the tilted thrust vector pushes the drone forward.
- Move backward: The two front motors spin faster than the rear motors, tilting the nose up, so the drone moves backward.
- Move left: The right-side motors spin faster than the left-side motors. This rolls the drone towards the left, and it slides/moves left.
- Move right: The left-side motors spin faster than the right-side motors, rolling the drone to the right so it moves right.
- Rotate (Yaw): The pair of CW motors and the pair of CCW motors are sped up/slowed down relative to each other. Since their spin reaction torques no longer cancel out, the whole drone body rotates left or right around its vertical axis, without changing its position much.

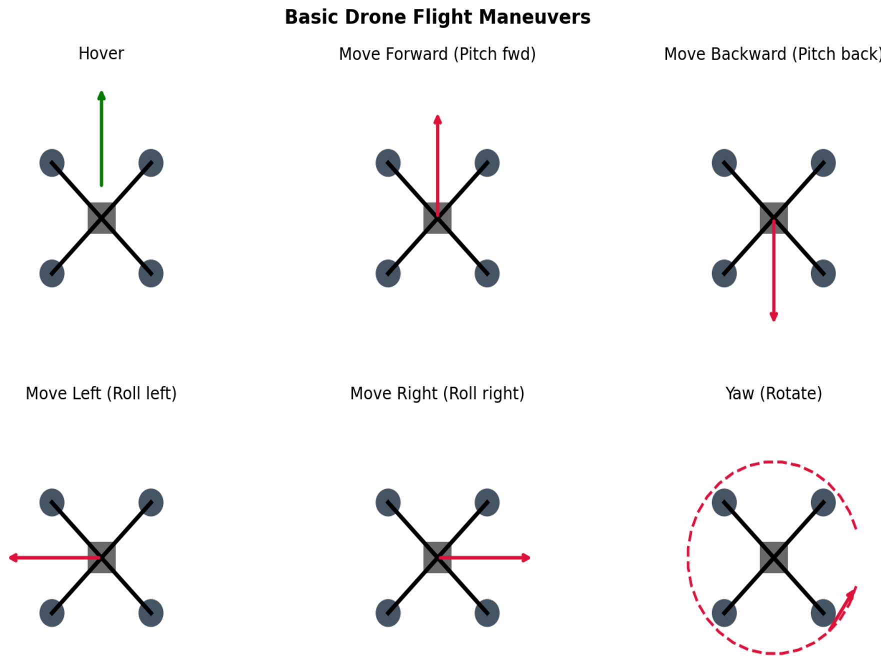

Figure 2: Basic drone flight maneuvers

Q6. Block diagram of a drone and function of each component. (10 Marks)

Ans:

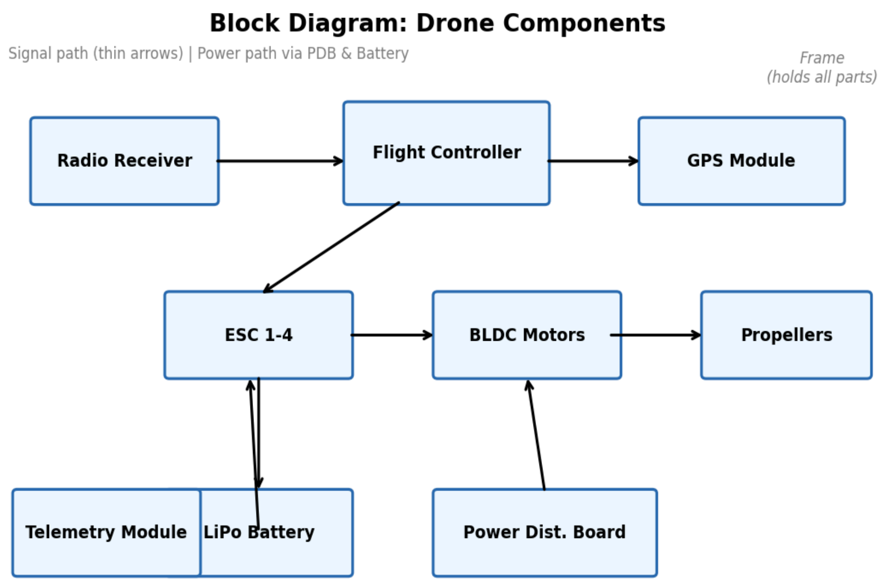

Figure 3: Block diagram of a typical quadcopter

Flight Controller (FC): The 'brain' of the drone. It is a small circuit board with a microcontroller and sensors (gyroscope, accelerometer). It reads pilot commands and sensor data, and decides how fast each motor should spin to keep the drone stable.

ESC (Electronic Speed Controller): Takes the low-power signal from the flight controller and converts it into the high-current three-phase power needed to drive each brushless motor at the correct speed.

Brushless (BLDC) Motors: Spin the propellers. They are preferred over brushed motors because they are more efficient, run cooler, and last longer.

Propellers: Push air downward to generate the lift/thrust that keeps the drone in the air.

Battery (LiPo): Stores and supplies electrical energy to power the motors, flight controller, and other electronics.

Power Distribution Board (PDB): Distributes battery power evenly to all four ESCs and other electronic components.

GPS Module: Gives the drone its exact location (latitude, longitude, altitude), enabling features like position hold, return-to-home, and waypoint navigation.

Radio Receiver: Receives the pilot's stick commands sent wirelessly from the radio transmitter and passes them to the flight controller.

Frame: The mechanical skeleton that holds all components together and gives the drone its shape and strength.

Telemetry Module: Sends live flight data (battery voltage, altitude, GPS position, speed) back to the ground station/pilot's screen.

Q7. Role of components. (10 Marks)

Ans: Flight Controller: Processes sensor data and pilot input to compute exact motor speeds needed for stable, controlled flight.

ESC: Converts DC battery power into the controlled three-phase AC signal that drives a BLDC motor at a commanded speed.

BLDC Motor: Provides the rotational power to spin the propeller and generate thrust.

Propeller: Converts the motor's rotation into thrust by pushing air downward.

Li-Po Battery: Main power source; lightweight and capable of delivering high current bursts needed by powerful motors.

GPS: Provides positioning data used for navigation, autonomous missions, and return-to-home safety features.

Radio Transmitter: The handheld remote control used by the pilot to send commands (throttle, roll, pitch, yaw) to the drone.

Frame: Structural base that holds motors, electronics, and battery together while resisting vibration and crashes.

Battery Eliminator Circuit (BEC): Steps down the main battery voltage to a lower, stable voltage (usually 5V) to safely power the flight controller, receiver, and other low-voltage electronics.

Telemetry Module: Provides two-way wireless data link between the drone and ground station, sending flight status and sometimes receiving mission commands.

Q8. Sensors: working principle and applications. (10 Marks)

Ans: IMU (Inertial Measurement Unit).

- Purpose: Measures the drone's orientation and motion.
- Working: Combines an accelerometer, gyroscope (and sometimes a magnetometer) into a single chip to measure linear acceleration and angular rotation rate.
- Application: Core sensor for stabilization in every flight controller.

Accelerometer.

- Purpose: Measures linear acceleration and tilt.
- Working: Uses tiny MEMS structures that shift under acceleration, changing capacitance, which is converted to an electrical signal.
- Application: Detects tilt angle to help keep the drone level.

Gyroscope.

- Purpose: Measures the rate of rotation (angular velocity).
- Working: A vibrating MEMS element experiences a Coriolis force when rotated, which is measured electronically.
- Application: Detects and corrects unwanted spinning/rotation instantly for stability.

Magnetometer (Compass).

- Purpose: Measures the earth's magnetic field to find heading direction.
- Working: Detects magnetic field strength along different axes to calculate compass heading.

- Application: Helps the drone know which direction is North for accurate navigation.

Barometer.

- Purpose: Measures altitude.
- Working: Senses atmospheric air pressure, which decreases predictably with height.
- Application: Used for altitude hold mode, keeping the drone at a constant height.

GPS.

- Purpose: Provides absolute global position.
- Working: Receives timing signals from multiple satellites and calculates position via triangulation.
- Application: Used for waypoint navigation, geofencing, and return-to-home.

Optical Flow Sensor.

- Purpose: Detects sideways drift, especially indoors.
- Working: A small downward-facing camera compares consecutive images to measure ground movement, similar to how an optical mouse works.

Ultrasonic Sensor.

- Purpose: Measures short-range distance to the ground or obstacles.
- Working: Sends out a sound pulse and measures the time taken for the echo to return.
- Application: Used for precise low-altitude height hold and obstacle avoidance.

LiDAR.

- Purpose: Creates precise 3D maps of surroundings.
- Working: Sends laser pulses and measures time-of-flight of reflected light to calculate distance.
- Application: Used in mapping, surveying, and advanced obstacle avoidance.

Camera.

- Purpose: Captures visual images/video.
- Working: Uses an image sensor (CMOS) to capture light and convert it into a digital image.
- Application: Used for aerial photography, inspection, and vision-based navigation.

Q9. Comparisons. (5 Marks)

Ans: (a) GPS vs Optical Flow

Point of Difference	GPS	Optical Flow
Works best	Outdoors, with clear open sky	Indoors or close to the ground
How it works	Satellite triangulation	Downward camera compares consecutive images
Accuracy	Good outdoors, poor indoors	Very accurate at low altitude, needs good lighting
Limitation	Blocked by roofs/tall buildings	Needs a textured surface below and enough light

(b) LiDAR vs Ultrasonic

Point of Difference	LiDAR	Ultrasonic
Signal used	Laser light	Sound waves
Range	Long range, high precision	Short range only
Cost & power use	Expensive, power-hungry	Cheap, low power
Best suited for	Detailed 3D mapping, surveying	Simple low-altitude height sensing

(c) Gyroscope vs Accelerometer

Point of Difference	Gyroscope	Accelerometer
Measures	Rate of rotation (angular velocity)	Linear acceleration and tilt (via gravity)
Speed of response	Very fast, but drifts over time	Slower, but no long-term drift
Weakness	Accumulates error (drift) over time	Noisy during vibration/fast movement
Used together as	Combined via sensor fusion for accurate, drift-free orientation estimate	Combined via sensor fusion for accurate, drift-free orientation estimate

Q10. 6 Degrees of Freedom (6DOF). (5 Marks)

Ans: 6DOF means a rigid body, like a drone, can move in six independent ways in 3D space: three types of straight-line movement (translation) and three types of rotation.

- Translation along X-axis: Moving forward or backward (surge).
- Translation along Y-axis: Moving left or right (sway).
- Translation along Z-axis: Moving up or down (heave).
- Rotation about X-axis: Roll - tilting left/right.
- Rotation about Y-axis: Pitch - tilting nose up/down.
- Rotation about Z-axis: Yaw - rotating left/right while facing a different direction.

A drone needs to control all six of these to fly freely and precisely, which is why it needs at least 4 independently controllable motors along with an IMU to sense its own motion in all these directions.

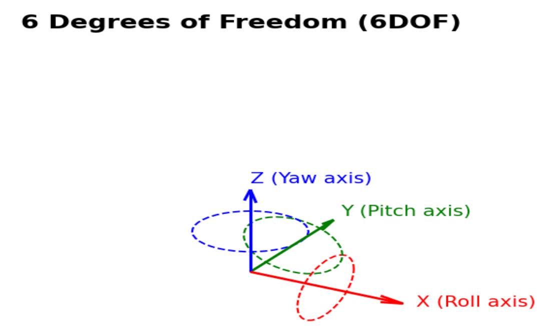

Figure 4: 6 Degrees of Freedom - 3 translations + 3 rotations

Q11. Roll, Pitch, Yaw, Surge, Sway, Heave with examples. (10 Marks)

Ans: Roll: Rotation about the front-to-back (X) axis.

Example: The drone tilts its left side down and right side up to move sideways to the left.

Pitch: Rotation about the left-to-right (Y) axis.

Example: The drone tilts its nose down to accelerate forward, like leaning forward on a bicycle.

Yaw: Rotation about the vertical (Z) axis.

Example: The drone spins in place to point its camera towards a different direction without changing location.

Surge: Straight-line movement along the X-axis (forward/backward).

Example: The drone flies straight ahead in a survey line over a field.

Sway: Straight-line movement along the Y-axis (left/right).

Example: The drone shifts sideways to frame a photo without changing which way it's facing.

Heave: Straight-line movement along the Z-axis (up/down).

Example: The drone climbs higher to clear a tree or descends to land.

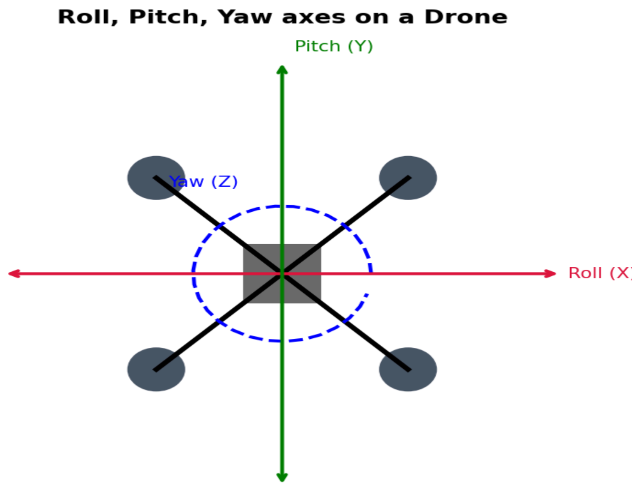

Figure 5: Roll, Pitch and Yaw axes on a drone

Q12. Complete signal flow in a drone. (5 Marks)

Ans: The complete chain of control in a drone works like this:

- 1. Pilot Input: The pilot moves the sticks on the remote (throttle, roll, pitch, yaw).
- 2. Radio Transmitter: Converts stick positions into a radio signal and transmits it wirelessly.
- 3. Radio Receiver: Mounted on the drone, it catches this radio signal and passes it on as electrical pulses.
- 4. Flight Controller: Reads the receiver's commands along with its own sensor data (IMU, GPS, barometer), and calculates the exact speed each individual motor should run at.
- 5. ESC: Receives the FC's motor-speed commands and converts them into the actual high-current three-phase power for each motor.
- 6. BLDC Motors: Spin at the commanded speed.
- 7. Propellers: Convert the motor's spin into thrust (air pushed downward).
- 8. Drone Movement: The combined thrust from all four propellers results in the drone's actual movement - hover, climb, tilt, or turn - exactly as the pilot intended.

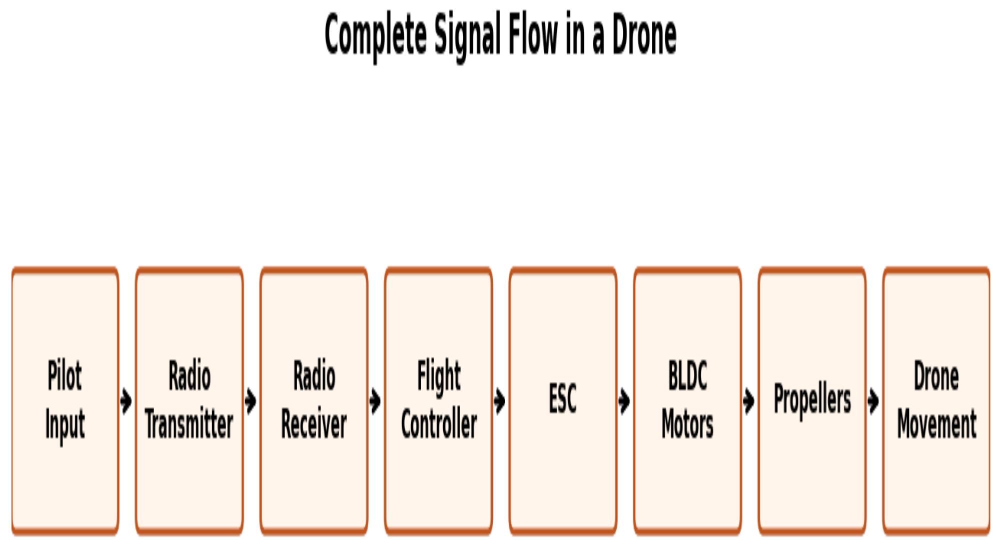

Figure 6: Complete signal flow, from pilot input to drone movement

Q13. Why PID control is important in drones. (5 Marks)

Ans: A drone is naturally unstable - even a light gust of wind can tilt it and make it flip over if left uncorrected. PID (Proportional-Integral-Derivative) control is the algorithm the flight controller uses to constantly correct the drone's angle and keep it stable, dozens or hundreds of times per second.

- Proportional (P): Looks at the current error (difference between desired angle and actual angle) and applies a correction proportional to how large that error is. Bigger tilt means a bigger, faster correction.
- Integral (I): Looks at the accumulated error over time. This corrects small, persistent errors that the P term alone cannot fully fix, such as a constant slight drift caused by an unevenly balanced frame.
- Derivative (D): Looks at how fast the error is changing, and applies a correction to prevent overshoot - like a brake that slows down the correction as the drone approaches the target angle, preventing oscillation/wobbling.

Together, P+I+D work continuously in a feedback loop (sensor reads tilt to controller to motors to sensor reads tilt again), which is what allows a drone to hover rock-steady in the air even in windy conditions.

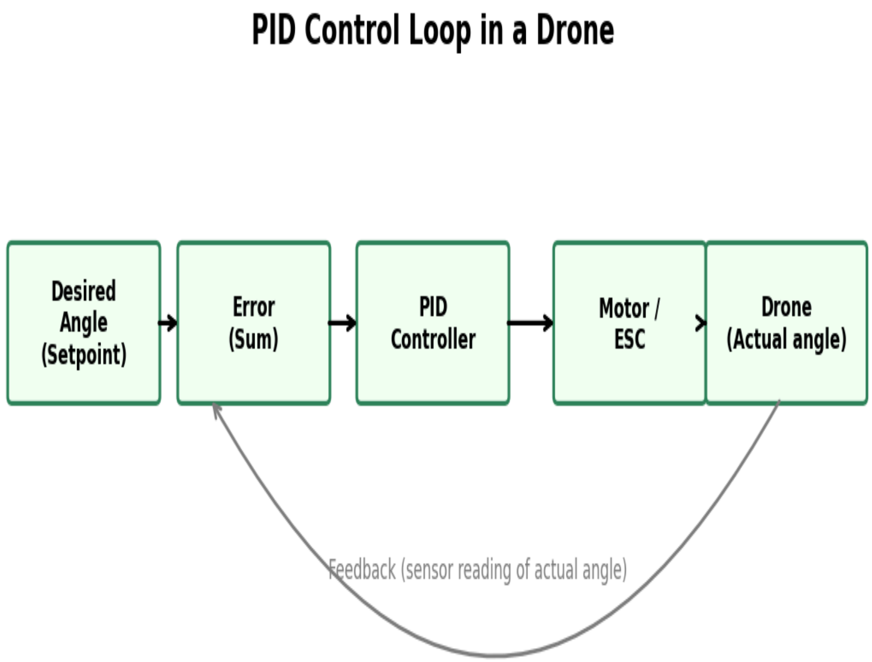

Figure 7: PID feedback control loop in a drone

Q14. Bonus - Research Questions (Any TWO). (10 Marks)

Ans: 1) Compare DJI, ArduPilot, and PX4 flight controllers

DJI, ArduPilot, and PX4 are the three most talked-about names in the drone flight-controller world, but they are quite different in nature - one is a closed commercial product, and the other two are open-source firmware projects.

Feature	DJI	ArduPilot	PX4
Type	Closed-source, proprietary hardware + software ecosystem	Open-source firmware (free to modify)	Open-source firmware (free to modify)
Ease of use	Very beginner-friendly, plug-and-play, polished mobile app	Moderate - needs some configuration (Mission Planner/QGroundControl)	Moderate - needs configuration (mostly QGroundControl)
Customization	Very limited, locked ecosystem	Highly customizable, supports many vehicle types (copter, plane, rover, sub)	Highly customizable, strong modular software architecture
Community / Support	Official DJI support only, no public source code	Very large hobbyist community, long history since 2010	Backed by Dronecode Foundation, strong in academia/research and industrial drones
Typical users	Hobbyists, photographers, commercial pilots wanting reliability out of the box	Hobbyists, researchers, and DIY builders	Researchers, universities, and companies building custom autonomous drones

Cost	Expensive but all-in-one (drone + FC + app included)	Low cost - runs on affordable boards like Pixhawk	Low cost - also runs on Pixhawk-family boards

In short: DJI is best if you just want a drone that works reliably without touching any code, while ArduPilot and PX4 are best for students, researchers, or companies who want full control over the flight software and are comfortable with some technical setup. ArduPilot is generally seen as slightly easier to start with, while PX4 has a more modern, modular software design favoured in academic and research projects.

2) How do drones return to home automatically?

Ans: Return-to-Home (RTH) is a safety feature that brings the drone back to its take-off point automatically, without the pilot's manual control. It works like this:

- Home point recording: The moment the drone gets a strong GPS lock (usually just before take-off), the flight controller saves that exact GPS coordinate as the 'home point'.
- Trigger: RTH can be triggered automatically (low battery, lost radio signal beyond a safe time limit) or manually by the pilot pressing an RTH button.
- Path planning: The flight controller calculates the direction and distance back to the home point using its current GPS position compared to the saved home coordinates.
- Climb to safe altitude: Many drones first climb to a pre-set safe height, so they don't crash into trees or buildings on the way back.
- Navigation: Using GPS and the compass (magnetometer) for heading, the drone flies in a straight line back towards the home point, correcting its path continuously using PID-controlled stabilization.
- Landing: Once above the home point, the drone either lands automatically by gradually reducing throttle, or simply hovers there and waits for the pilot to take back manual control.

This entire process depends heavily on a healthy GPS signal - which is why RTH is far less reliable indoors or in GPS-denied areas.

Q15. Practical Activity - Labeling Every Component of a Quadcopter Drone. (10 Marks)

Ans: For this activity, a labeled diagram of a typical quadcopter has been drawn below, with each major component numbered and described in the table.

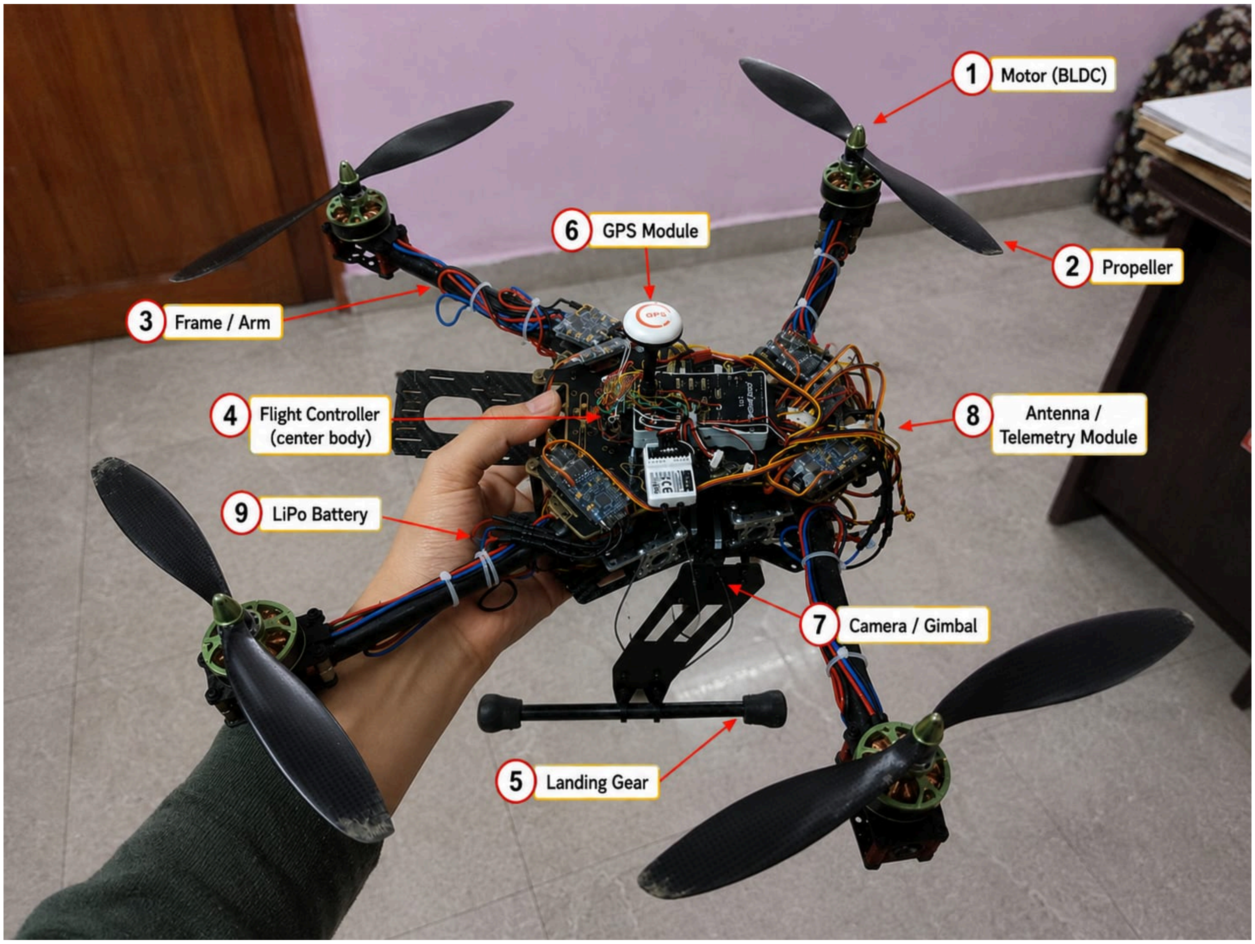

Figure 8: Labeled diagram of a quadcopter drone

Label	Component	Function
1	Motor (BLDC)	Spins the propeller to generate thrust
2	Propeller	Pushes air downward to create lift
3	Frame / Arm	Structural support connecting motors to the central body
4	Flight Controller (center body)	Processes sensor data and controls all motor speeds
5	Landing Gear	Legs that support the drone on the ground during take-off and landing
6	GPS Module	Provides position data for navigation and return-to-home
7	Camera / Gimbal	Captures photos/video and provides visual feedback
8	Antenna / Telemetry Module	Sends and receives wireless data with the ground station
9	LiPo Battery	Supplies power to all electronics and motors

Observation: Every component has a clear, specific job, but they all depend on each other in a tight loop - sensors feed data to the flight controller, the flight controller commands the motors through the ESCs, and the resulting movement is sensed again by the same sensors. This closed feedback loop is what makes a drone fly in a stable, controlled way rather than just spinning propellers randomly.

Reflect and Answer (In Own Words)

These are my own personal reflections after learning about drones in this course, written in simple language based on what I understood.

1. Why do quadcopters require four motors instead of one?

Ans: A single motor can only spin one propeller, and one propeller alone cannot control direction - it can only push a drone up or down, not tilt it or turn it. With four motors placed at four different corners, the drone can speed up or slow down each one individually, which lets it tilt forward, backward, sideways, or rotate. Basically, one motor gives lift but no control, while four motors together give both lift and full directional control.

2. What would happen if one motor stops working during flight?

Ans: If it's a normal quadcopter, losing one motor is usually fatal for the flight - the remaining three motors create an unbalanced thrust, the drone starts spinning uncontrollably, and it crashes because there's no way to cancel out the torque from the opposite working motors anymore. This is exactly why hexacopters and octocopters exist - with 6 or 8 motors, losing one still leaves enough motors to balance the drone and land safely.

3. Why are brushless motors preferred over brushed motors?

Ans: Brushed motors have physical carbon brushes that rub against a commutator to pass current, and this constant friction wears them out fast, wastes energy as heat, and limits their speed. Brushless motors use electronic switching (through the ESC) instead of physical contact, so there's way less friction, they run cooler, last longer, and can spin much faster - which is exactly what's needed to lift a drone efficiently on limited battery power.

4. Can a drone fly indoors without GPS? Explain.

Ans: Yes, it can, but it needs a different way to know its position since GPS signals don't really work well indoors (the satellite signal gets blocked by the roof and walls). Indoors, drones usually rely on an optical flow sensor (a small downward camera that tracks how the ground is moving under it) along with ultrasonic sensors for height, and sometimes IMU data alone for short flights. So GPS is not compulsory for flight - it's just one way to know position, and indoor drones substitute it with vision-based or sensor-based positioning instead.

5. Why are Li-Po batteries used in drones instead of standard batteries?

Ans: Drones need a battery that is light in weight but can still push out a huge burst of current instantly, since spinning four motors at high speed draws a lot of power. Standard batteries like alkaline or even regular Li-ion cells are either too heavy for their capacity or can't discharge current fast enough. Li-Po (Lithium Polymer) batteries are lightweight, can be shaped flat to fit the drone body, and can discharge very high current quickly, which is why they became the standard choice.

6. What factors reduce a drone's flight time?

Ans:

- Heavier payload (extra camera, sensors, cargo) makes the motors work harder and drain the battery faster.
- Flying in strong wind makes the drone constantly correct its position, using more power than calm-air hovering.
- Aggressive flying (fast acceleration, sharp turns, quick climbs) uses more current than smooth, steady flight.
- Old or low-quality batteries hold less charge and degrade the available flight time over repeated use.
- Cold temperatures reduce battery efficiency and chemical performance, shortening flight time.

7. How can drone stability be improved during strong winds?

Ans: The flight controller can be tuned with more aggressive PID values so it reacts faster to sudden tilts caused by gusts. Using a heavier or more aerodynamic frame also helps the drone resist being pushed around as easily. Adding more powerful motors gives extra thrust margin to fight against the wind instead of just barely staying stable. And practically speaking, flying at a lower altitude, closer to the ground, usually means less wind exposure compared to flying high up.

8. Why is sensor fusion important in drones?

Ans: No single sensor gives a perfect picture on its own - a gyroscope is fast and responsive but drifts and builds up error over time, while an accelerometer can sense absolute tilt using gravity but gets noisy and shaky during vibration or fast movement. Sensor fusion means combining data from multiple sensors (like a Kalman filter or complementary filter combining gyroscope + accelerometer + magnetometer) so their weaknesses cancel out and their strengths add up. This gives the drone one accurate, stable estimate of its real orientation, which is essential for it to stay balanced in the air.

9. Explain how GPS and IMU work together for stable navigation.

Ans: GPS tells the drone where it is in the world (its absolute latitude, longitude, and altitude), but it updates fairly slowly (once every second or so) and can be a little noisy. The IMU (gyroscope + accelerometer) updates extremely fast (hundreds of times per second) and tells the drone how it's moving and tilting right now, but it can't tell absolute position on its own and drifts over time. Together, the flight controller uses the IMU for instant, moment-to-moment stability corrections, while GPS is used every second or so to correct and confirm the drone's actual position, canceling out any drift the IMU alone would build up. This combination is what lets a drone hold a steady position outdoors even in light wind.

10. Designing a drone for mountain rescue operations - which sensors/components and why?

Ans: For a mountain rescue drone, I would prioritize range, reliability in bad weather, and the ability to spot people who might not be easily visible. My choices would be:

- Thermal camera - to detect body heat of a stranded or injured person even in fog, snow, or at night when a normal camera would fail.
- LiDAR - to accurately map rocky, uneven mountain terrain and detect obstacles like cliffs or trees, which is more reliable than ultrasonic at longer range.
- GPS + barometer combo - to track exact 3D position and altitude, which is critical in mountains where altitude changes constantly and needs to be reported accurately to the rescue team.

- Long-range telemetry module - so the ground team can keep contact with the drone over the large distances typical of mountain terrain.
- Fixed-wing or hybrid VTOL frame instead of a pure multirotor - because fixed-wing drones can cover much longer distances and stay airborne longer, which matters a lot in a large search area, while still being able to hover briefly over a found location if it's a hybrid design.
- Bigger, higher-capacity Li-Po battery or even a hybrid fuel-electric system - to maximize flight time for a wide-area search.

Overall, I'd trade off small-space maneuverability (which a small quadcopter is great at) for range, endurance, and all-weather sensing, since a mountain rescue mission is mostly about covering a large, harsh area quickly and reliably finding someone rather than fine indoor control.

--- End of Assignment ---